

Basic Principles of Medicine 1

Module : MSK

Lecture No: # 26

Lecture Title: Triplet Repeat Disorders and Pharmacogenetics – Clicker Questions and Discussions based on DLAs and assigned readings for genetic cases including muscular dystrophy, osteogenesis imperfecta, neurofibromatosis, triplet repeat expansion disorders, and pharmacogenetics- Clicker Questions and Discussions

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Required Reading

- Preview the video cases that are posted on Panopto page for SGU
- Triplet repeat expansion videos:
 - 01 General features – unstable repeat expansions
 - 02 Fragile X syndrome
 - 03 Huntington disease
 - 04 Myotonic dystrophy
 - 05 Friedreich ataxia
 - 05 The SCA disorders
- Pharmacogenetic videos
 - 01 pharmacogenetic concepts
 - 02 Malignant hyperthermia
 - 03 G6PD deficiency
 - (we will advise you to review this in CPR module, before L5)
 - 04 TPMT polymorphism (done in FTM module)

Recommended Prereadings: From Korf and Irons *Human Genetics and Genomics*, 4th edition:

The following pages from the Korf and Irons *Human Genetics and Genomics* book

- A. In chapter 3 (Patterns of Inheritance) read the section for Triplet Repeat Expansion Disorders on pages 54 – 57 (See also Figure 3.28 which is on page 58).
 - a. Note that the description for Friedreich ataxia on page 57 has an error. The mutation causing Friedreich ataxia leads to loss of expression of the gene (local formation of heterochromatin, and loss of expression). The book erroneously states that the triplet repeat causes abnormal RNA processing
- B. Chapter 3, see Hot Topic 3.1 on 56 and 57 – for FXTAS – which describes how, in the case of fragile X, the premutation allele has a disorder called FXTAS which is associated with **overexpression** of the *FMR1* gene leading to premature ovarian insufficiency and ataxia. While fragile X syndrome itself is caused by **loss** of expression of *FMR1* and defects in synaptic maintenance.
- C. Chapter 4 (The Human Genome) see Clinical Snapshot 4.1 – Huntington disease on page 74
- D. Chapter 16 (Pharmacogenetics) on pages 230 – 237. Note that this is a very short chapter which focuses on the case of a boy who develops malignant hyperthermia during tonsillectomy surgery.
 - a. Be sure to read pages 234 – 235 which provides a few other classic examples of pharmacogenetic interactions including the action of TPMT enzyme, the conversion of Codeine to Morphine by the enzyme debrisoquinone hydroxylase (encoded by the CYP2D6 gene), and the metabolism of Warfarin.
 - i. Note that the legend for figure 16.3 has an error: the action of the Thiopurine methyltransferase (TPMT) enzyme is correctly described in the Panopto video, and its attached slides. Briefly: Mercaptopurine is converted into compounds that will inhibit synthesis of purines; this reduces the rate of cellular proliferation. TPMT stops this from occurring by deactivating the thiopurine (6-mercaptopurine). The Figure legend in the book erroneously states that the TPMT changes how thiopurine is incorporated into DNA.

Pre-Lecture - Required Reading and Objectives

- DLA associated with flipped classroom
- DLA Title: Pharmacogenetics
- DLA Title: Triplet Repeat Disorders

Learning Objectives:

SOM.MKIII.BPM1.2.MSK.1.GNET.0166	Explain how repetitive sequences show instability such that in some individuals the size of the repeat expands in the germ line
SOM.MKIII.BPM1.2.MSK.1.GNET.0167	Explain how most triple repeat disorders show decreased age of onset and increased severity when the repeat size expands and how parental origin may sometimes influence expansions and therefore anticipation
SOM.MKIII.BPM1.2.MSK.1.GNET.0168	Describe fragile X syndrome
SOM.MKIII.BPM1.2.MSK.1.GNET.0169	Explain how fragile X syndrome is a disorder caused by a loss of expression of a gene
SOM.MKIII.BPM1.2.MSK.1.GNET.0170	Describe fragile X-associated tremor ataxia syndrome and how it is caused by an overexpression of a gene
SOM.MKIII.BPM1.2.MSK.1.GNET.0171	Describe Huntington disease (HD): how it may be diagnosed, both symptomatic and presymptomatic, via a genetic test; therapeutic use of antisense oligonucleotides
SOM.MKIII.BPM1.2.MSK.1.GNET.0172	Describe the genetics and clinical features of myotonic dystrophy
SOM.MKIII.BPM1.2.MSK.1.GNET.0173	Describe the genetics and clinical features of Friedrich ataxia and spinocerebellar ataxia

Objectives: Lecture

SOM.MK.III.BPM1.2.MSK.1.GNET.0174	Outline the classic examples of gene-drug interactions
SOM.MK.III.BPM1.2.MSK.1.GNET.0155	Analyze the genetic principles of triple repeat disorders. Review concepts and importance of pharmacogenetics and personalized medicine.

Clicker Questions Next!

Please prepare to answer
the clicker questions
on the slides.

A 70-year-old woman is brought to the physician because of a slow progression of apparent cognitive decline. She has good memory of events from long ago, but her memory of events from just a few hours ago is compromised. MRI scan shows mild cortical atrophy. She is diagnosed with Alzheimer disease. Provided that this diagnosis is correct, which of the following would be histologically observed in the brain of this patient upon autopsy?

- A. A protein with a polyglutamine tract expansion 0%
- B. Chromosomes containing fragile sites 0%
- C. A toxic mRNA 0%
-  D. Neurofibrillary tangle 0%
- E. Deficiency of the dystrophin (DMD) protein 0%

The major risk factor for Alzheimer disease is age.

- Some genetic modifier alleles may change this risk such as a specific *APOE* allele (recall Multifactorial inheritance lecture)
- Some autosomal dominant forms of Alzheimer disease are known
- By far..., most cases of Alzheimer disease are considered to be multifactorial in origin (and idiopathic)
- See page 94 of Korf and Irons, 4th edition for brief description of Alzheimer disease.

A 58-year-old man was prescribed the standard dose of the beta-blocker Metoprolol as an antihypertensive drug. He did not respond to the intervention. Which of the genes is most likely found in multiple copies (amplified) to explain the lack of clinical efficacy in this patient?

- | | |
|--------------------|----|
| A. <i>TPMT</i> | 0% |
| B. <i>CFTR</i> | 0% |
| C. <i>TP53</i> | 0% |
| 😊 D. <i>CYP2D6</i> | 0% |
| E. <i>G6PD</i> | 0% |
| F. <i>COL1A1</i> | 0% |
| G. <i>RYR1</i> | 0% |
| H. <i>NF1</i> | 0% |

Ultrarapid Metabolizers

- Ultrarapid metabolizers may require very high doses of drug such as **metoprolol**.
 - This is because metoprolol is metabolized and deactivated by CYP2D6.
- But an ultrarapid metabolizer might overdose with **codeine**, suffering respiratory depression or even respiratory arrest in response to standard doses.
 - This is because codeine is converted to its active form (morphine) by CYP2D6.

A poor metabolizer

- Could have an adverse reaction from a standard Metoprolol dose
- Look back and see how this is reversed for Codeine

A 32-year-old woman comes to the physician because of amenorrhea and ataxia. Her older brother also has ataxia, and her son has fragile X syndrome. Genetic test shows that she has the fragile X premutation, and she is diagnosed with FXTAS. Which of the following best explains her pathology?

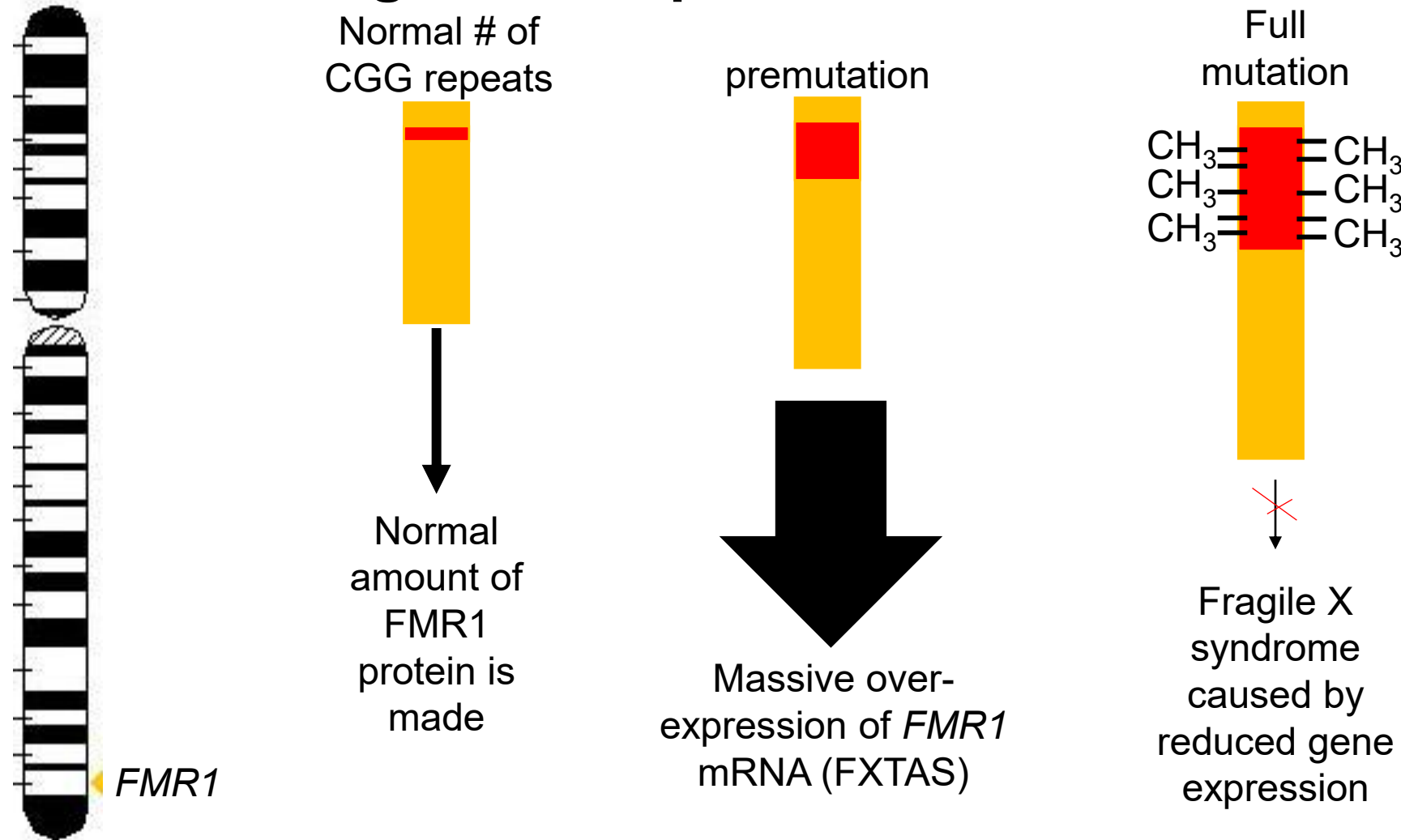
- | | |
|--|----|
| A. Loss of expression of the <i>FMR1</i> gene | 0% |
| 😊 B. Overexpression of the <i>FMR1</i> gene | 0% |
| C. Polyglutamine expansion in FMR1 protein | 0% |
| D. Oxidative damage to her red blood cells | 0% |
| E. Massive calcium release from the sarcoplasm | 0% |
| F. Formation of a fragile site on the X chromosome | 0% |

A 4-year-old boy is brought to the physician because of developmental delay and autistic behaviors. Genetic testing shows that he has a pathogenic expansion of 500 CGG repeats in the *FMR1* gene and he is diagnosed with fragile X syndrome. Which of the following best explains the condition in this child?

- A. His X chromosome has a fragile site 0%
- B. The *FMR1* gene is overexpressed 0%
-  C. The *FMR1* gene is not expressed 0%
- D. A polyglutamine expansion in the FMR1 protein occurs 0%
- E. FMR1 protein induces oxidative damage to red blood cells 0%
- F. FMR1 protein causes calcium release from the sarcoplasm 0%

- In patients with fragile X syndrome, the CGG expansion is so great, that *FMR1* gene expression is abolished
 - This happens because of cytosine methylation, and epigenetic gene silencing.

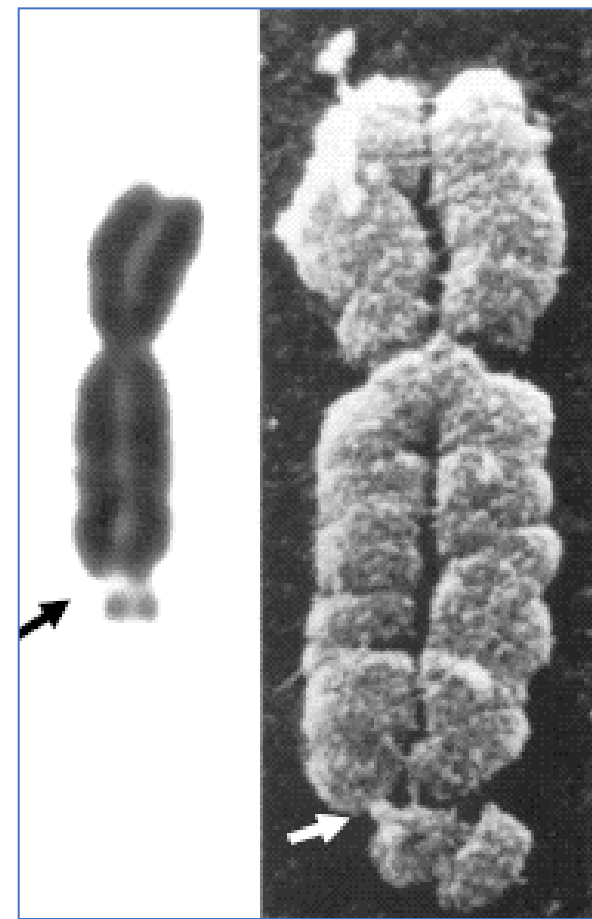
The *FMR1* gene: 3 important variations



In Chapter 3 of Korf and Irons textbook: see Hot Topic 3.1 on 56 and 57 – for FXTAS

- Normally, appropriate amounts of *FMR1* mRNA is made; so appropriate amounts of FMR1 protein is made, and synaptic connections are maintained
- In premutation alleles (intermediate CGG repeat lengths) – there is massive overexpression of the *FMR1* gene which causes FXTAS.
 - This is mysterious, and suggests that the mRNA has some kind of “toxic” effect
- In patients with fragile X syndrome, the CGG expansion becomes large enough that that cytosine bases become methylated
 - Methylation of the *FMR1* gene silences gene expression
 - This is an example of epigenetic gene silencing.

- Fragile site is revealed by a cytogenetic test: When cells are cultured in a folate deficient medium the X chromosomes might show a '*fragile*' site
 - Fragile site is seen only in patients who carry the full mutation
- In patients, the X chromosome is not really fragile, the disease is caused by reduced expression of the *FMR1* gene
- FRX is diagnosed by separation of PCR products by size.



The unusual appearance of the metaphase X-chromosome in cells cultured under folate limiting conditions gives the name to the disease.

The Fragile Site Does Not Explain The Disease Mechanism

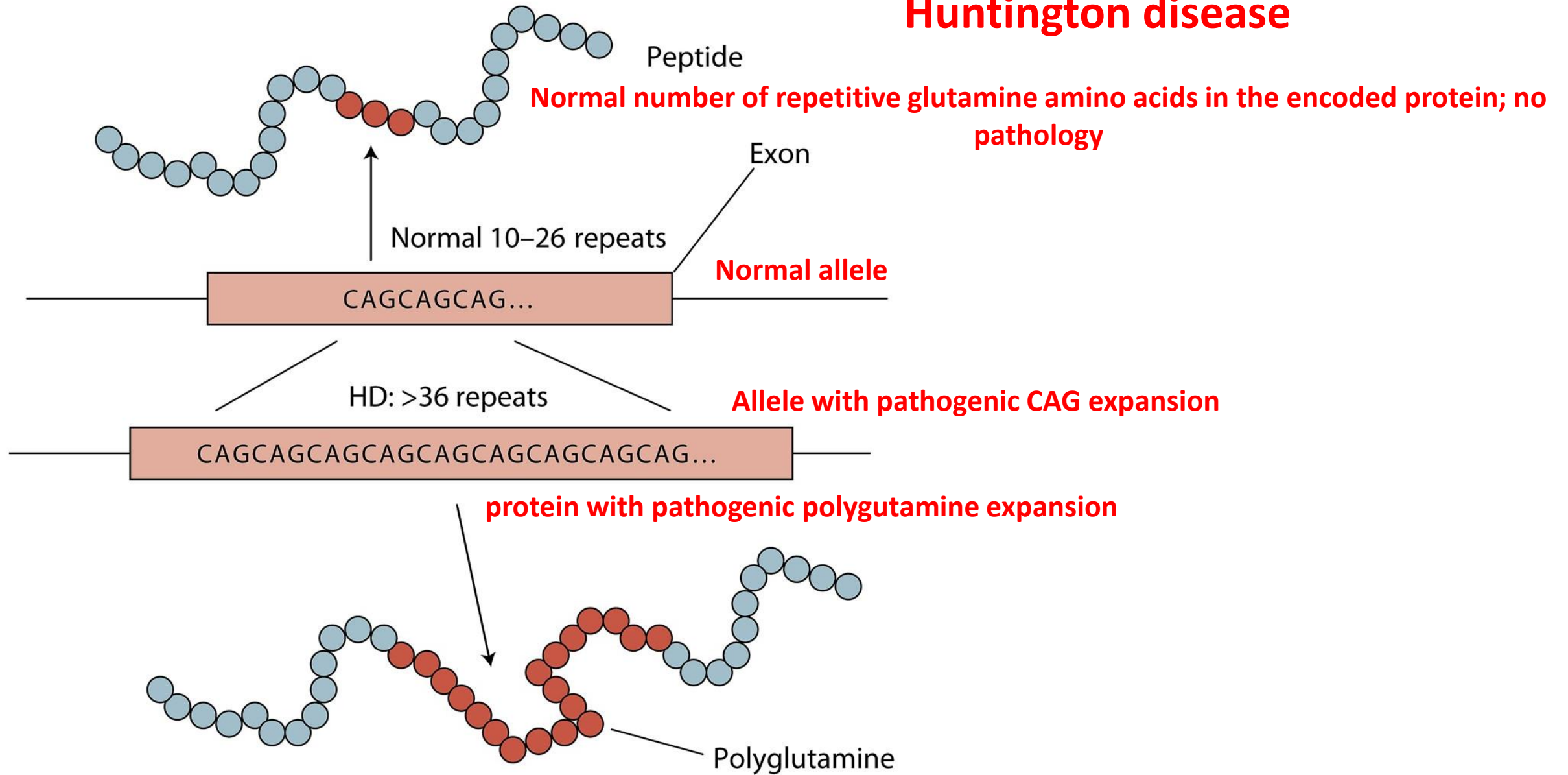
A 45-year-old woman comes to the physician after developing chorea. She is also exhibiting cognitive signs such as memory loss, sleep disturbances, and paranoia. Genetic testing shows that she has a triplet repeat expansion that explains her pathology. Where is this expansion most likely to be found in the affected gene?

- | | |
|------------------------|----|
| A. In the promoter | 0% |
| B. In the 3' UTR | 0% |
| 😊 C. In an exon | 0% |
| D. In an intron | 0% |
| E. In the 5' UTR | 0% |
| F. In the poly(A) tail | 0% |

A 39-year-old man develops loss of truncal muscle control and hyperkinetic writhing of his extremities (chorea). Huntington disease is suspected. Which of the following would most likely be used as part of the genetic test that would diagnose this patient?

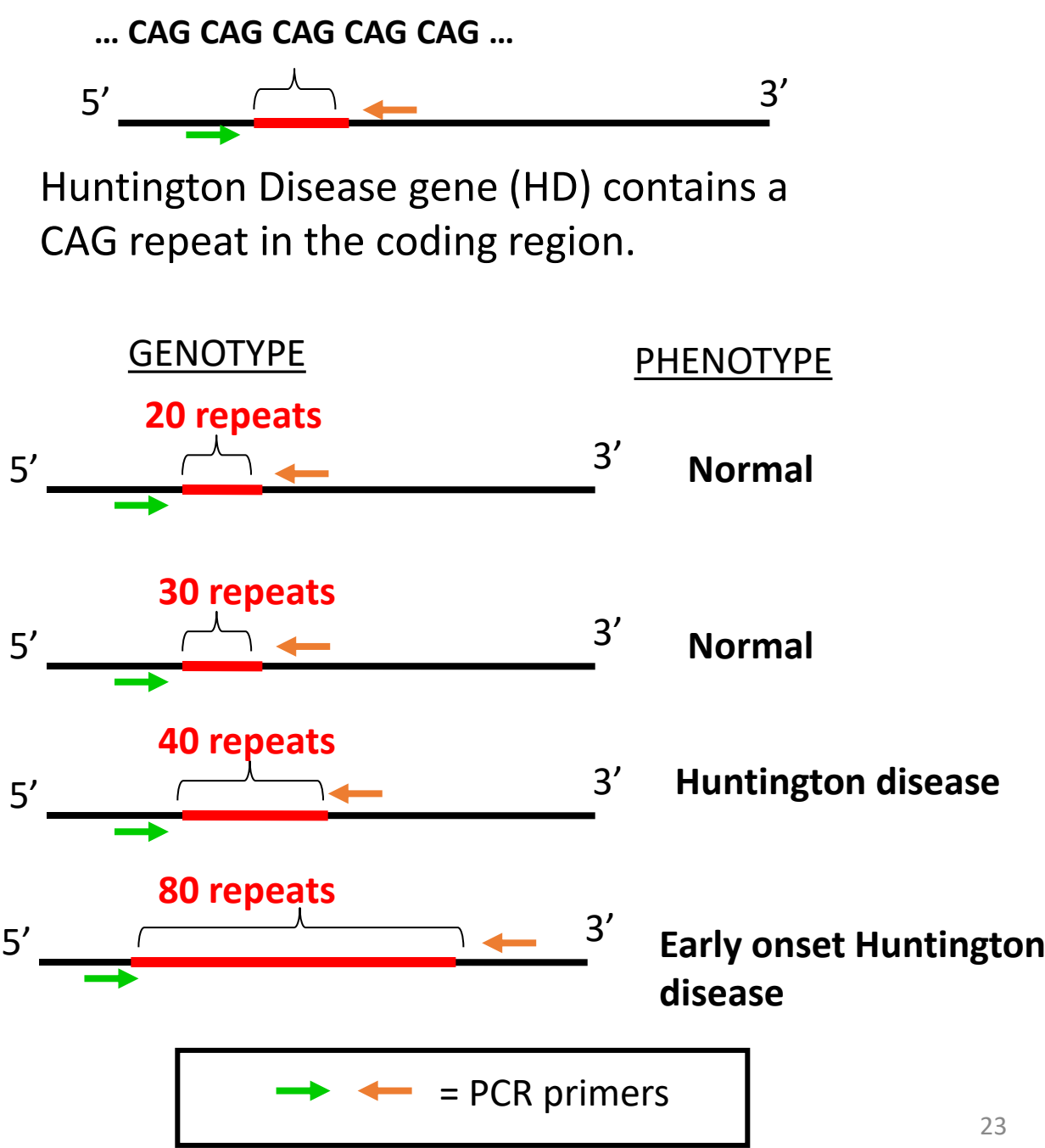
- | | |
|--------------|----|
| A. ASO blot | 0% |
| 😊 B. PCR | 0% |
| C. Array CGH | 0% |
| D. Karyotype | 0% |
| E. FISH | 0% |
| F. SNP Chip | 0% |

Huntington disease

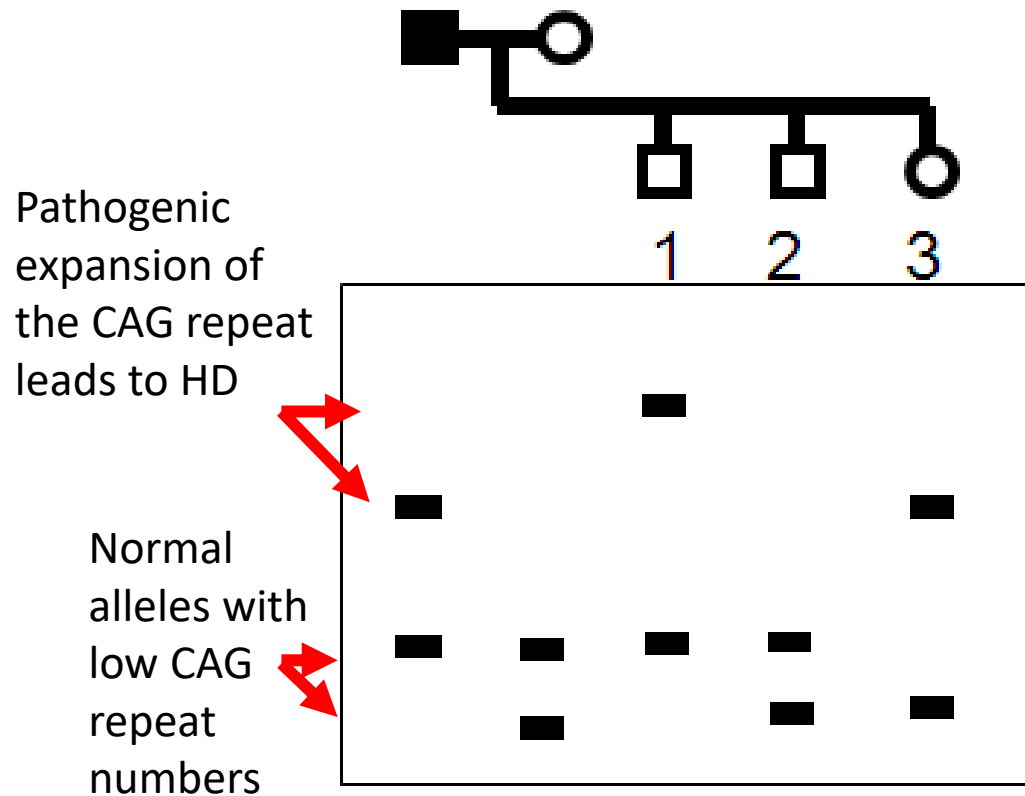


PCR may be used to amplify the triplet repeat expansions as occurs in **Huntington disease (the CAG repeat translates to a string of glutamine amino acids)**.

Primers flanking the repeat region will create different sized PCR products



Can separate PCR fragments by gel electrophoresis



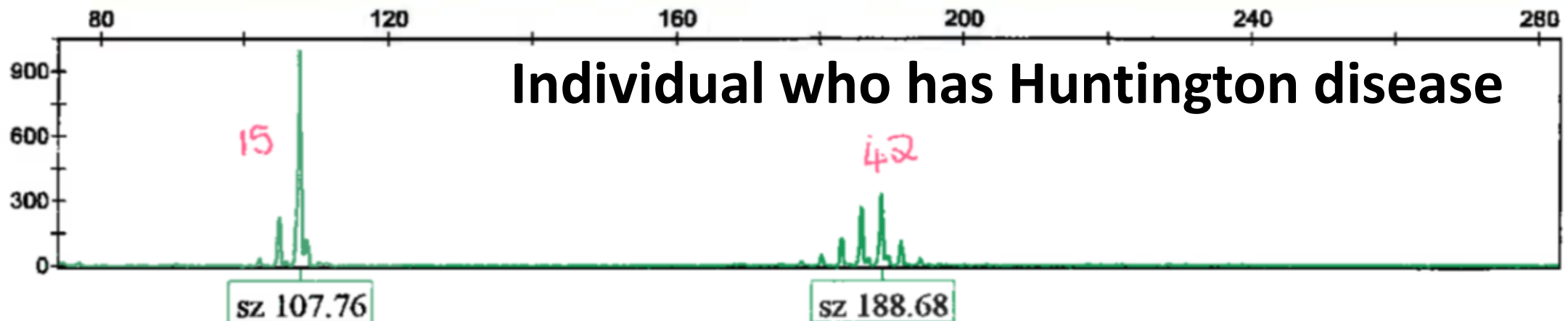
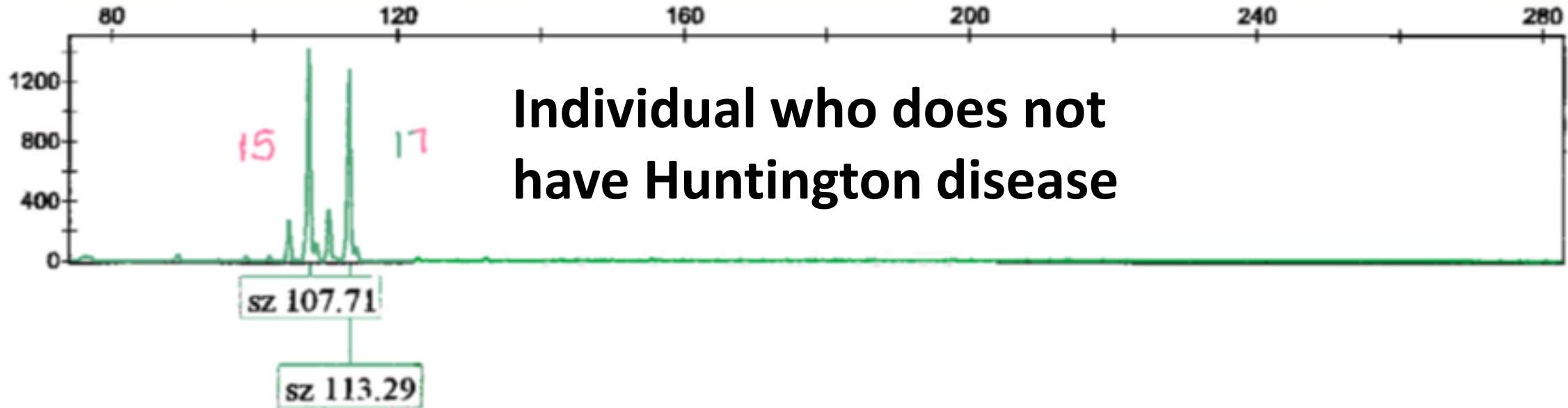
PCR

- Direct staining of PCR generated product
- Staining with ethidium bromide (DNA stain)

Note: Southern blot can reveal these same data:

- For Southern, genomic DNA is chopped up with a restriction enzyme, separated on a gel, denatured, then transferred to a membrane, and probed to reveal an area of interest
- This is complex, takes a lot of steps, and is messy

The clinical test uses column chromatography to separate PCR fragments by size – with internal controls, this is more precise.

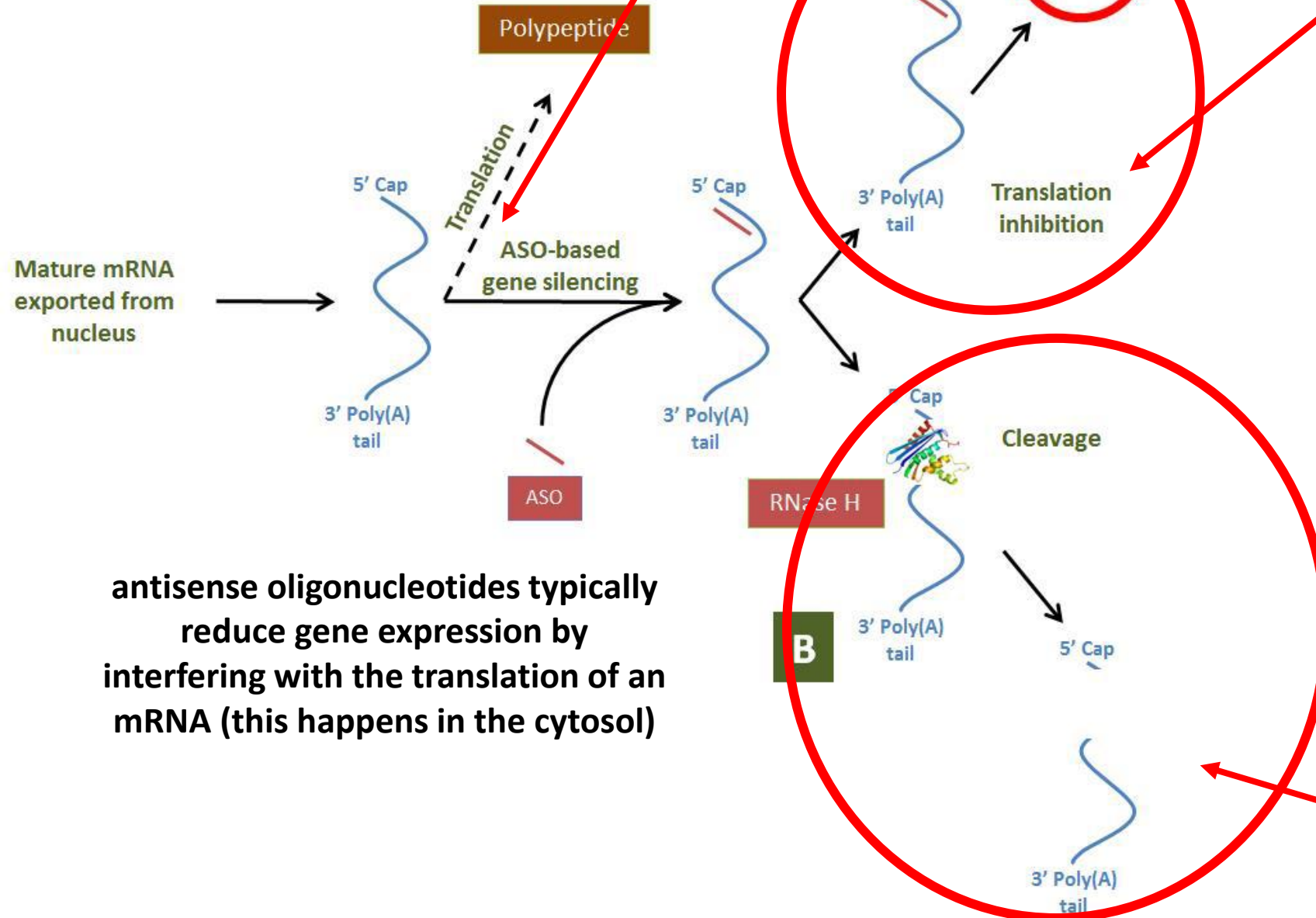


A 24-year-old woman enrolls in a research study after learning that her mother is affected with Huntington disease. Genetic testing shows that both her, and her mother have a pathogenic CAG expansion in the HTT gene. She receives intrathecal injections of an antisense oligonucleotide and is periodically monitored for CSF levels of the huntingtin protein. In which cellular compartment would this intervention most likely exert its effect?

- | | |
|----------------------|----|
| A. Lysosome | 0% |
| B. Nucleus | 0% |
| C. Smooth ER | 0% |
| 😊 D. Cytoplasm | 0% |
| E. Mitochondria | 0% |
| F. Golgi body | 0% |
| G. Secretory vesicle | 0% |

Here, ASO means antisense oligonucleotide

The ASO may block access of the target mRNA to the ribosome



- Gene silencing is not really the best way to describe this for a treatment of a disorder in humans.

- For HD, better to use the term HD-lowering

- Because some expression still occurs

antisense oligonucleotides typically reduce gene expression by interfering with the translation of an mRNA (this happens in the cytosol)

The ASO increase degradation of the target mRNA

A 32-year-old woman comes to the physician because she has a baby with severe hypotonia. The mom has similar symptoms, but not as severe, and the baby's grandmother is very mildly affected. This woman has difficulty releasing her grip after a firm handshake (myotonia). Genetic test shows a triplet repeat expansion in the *DMPK* gene. Which of the following best explains the variable expressivity in this family?

- A. Locus heterogeneity 0%
- B. Genetic drift 0%
- C. Mitochondrial inheritance 0%
- 😊 D. Anticipation 0%
- E. Frameshift mutation 0%

A 32-year-old woman comes to the physician to discuss her baby who has severe hypotonia. The mom has similar symptoms, but not as severe, and the baby's grandmother is very mildly affected. This woman has difficulty releasing her grip after a firm handshake (myotonia). Genetic test shows a triplet repeat expansion in the *DMPK* gene. Which of the following is most likely the molecular cause of the pathology in the affected individuals in this family?

- A. Loss of *DMPK* gene expression 0%
- 😊 B. Global change in mRNA processing 0%
- C. A polyglutamine tract expansion 0%
- D. Massive Ca^{++} release from the sarcomere 0%
- E. Reduced Cl^- transport across a membrane 0%
- F. Formation of heterochromatin 0%

A 66-year-old man undergoes surgery for knee replacement and has an adverse reaction to the anesthesia (succinylcholine and halothane). The patient is stabilized, and later, a genetic test shows that the man has an autosomal dominant mutation in the *RYR1* gene which encodes the ryanodine receptor. Which of the following lab test results would most likely be found in this patient directly following the adverse drug reaction?

- | | |
|---|----|
| A. Elevated serum ammonia | 0% |
| B. Elevated sweat chloride | 0% |
| C. Depressed serum blood urea nitrogen | 0% |
| D. Lack of glutathione (GSH) in the erythrocyte | 0% |
| 😊 E. Elevated serum creatine kinase | 0% |

A 5-year-old boy begins to show signs of ataxia and muscle weakness. A genetic test shows that he is homozygous for an expanded triplet repeat in an intron of a gene. Laboratory tests from a muscle biopsy show accumulation of oxidative damage, and excessive free iron in the mitochondria. Which of the following disorders is the most likely diagnosis?

- | | |
|---|----|
| A. Spinocerebellar ataxia | 0% |
| B. Myotonic dystrophy | 0% |
| 😊 C. Friedreich ataxia | 0% |
| D. Becker muscular dystrophy | 0% |
| E. Huntington disease | 0% |
| F. Fragile X tremor and ataxia syndrome | 0% |

Friedreich ataxia

- Autosomal recessive; childhood onset
- Gene is on an autosome
- Pathogenic triplet repeat expansion causes local formation of heterochromatin – so reduced gene expression
- Both parents are typically carriers (and unaffected)
- So,... anticipation is not observed in pedigrees

A 39-year-old man comes to the physician because of progressively worsening ataxia and balance problems. His father developed similar symptoms in his sixties. Genetic testing shows a pathogenic alteration of triplet repeats in the *ATXN3* gene. Which of the following would be a consequence of the mutation in this patient?

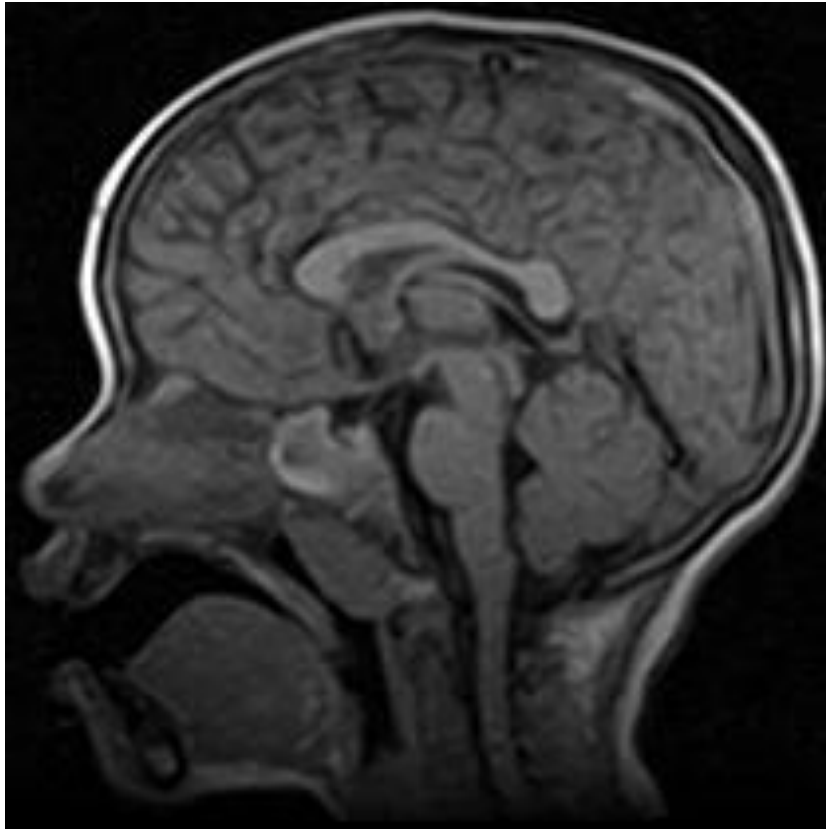
- 😊 A. Polyglutamine tract expansion 0%
- B. Methylation of the promoter 0%
- C. Conversion to heterochromatin 0%
- D. Inappropriate binding to splicing factors 0%
- E. Formation of a fusion protein 0%

Many of the SCAs (spinocerebellar ataxia) disorders are caused by CAG repeat expansions in genes that are expressed in the brain

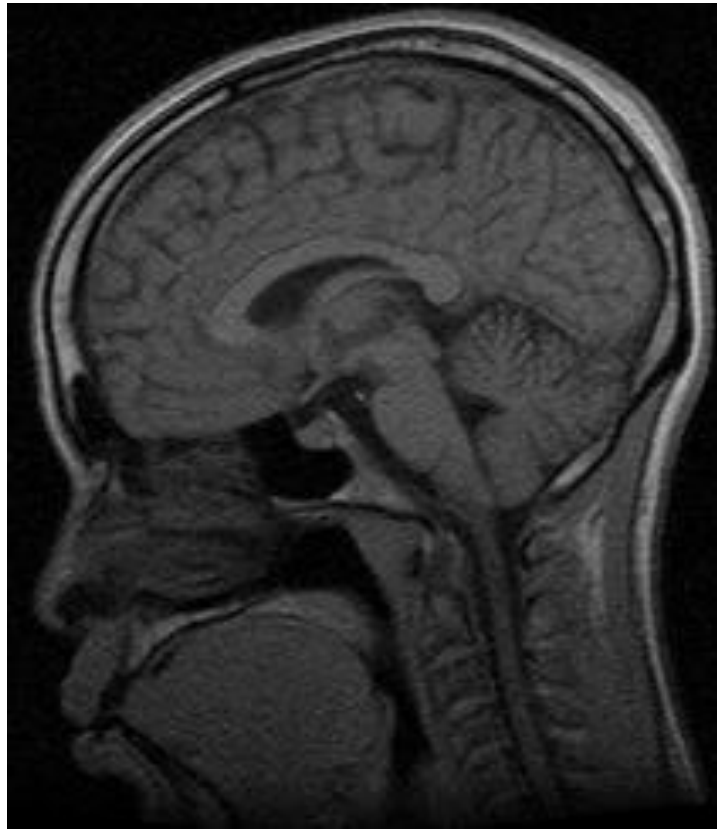
- The CAG repeat is a codon for glutamine
- Creates an expanded polyglutamine tract in the encoded protein
- Leads to 'gain of function' of the encoded protein
- Loss of neuronal tissue in cells that express that gene → protein

Brain MRI of an individual with spinocerebellar ataxia 3 shows progressive degeneration of cerebellar structures

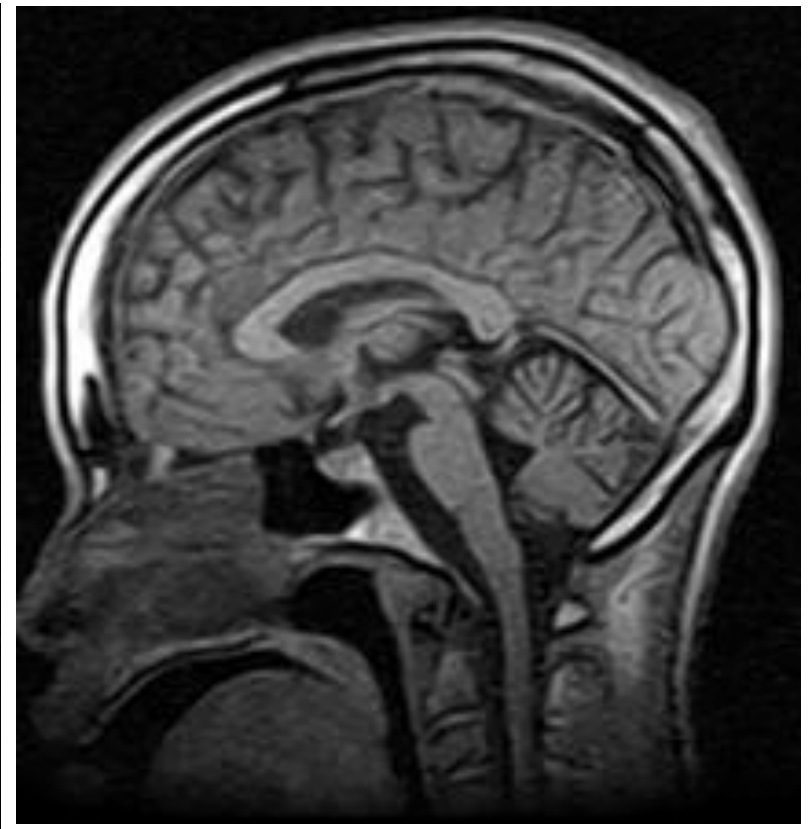
Before symptom onset:
Pons and cerebellar
structures are normal



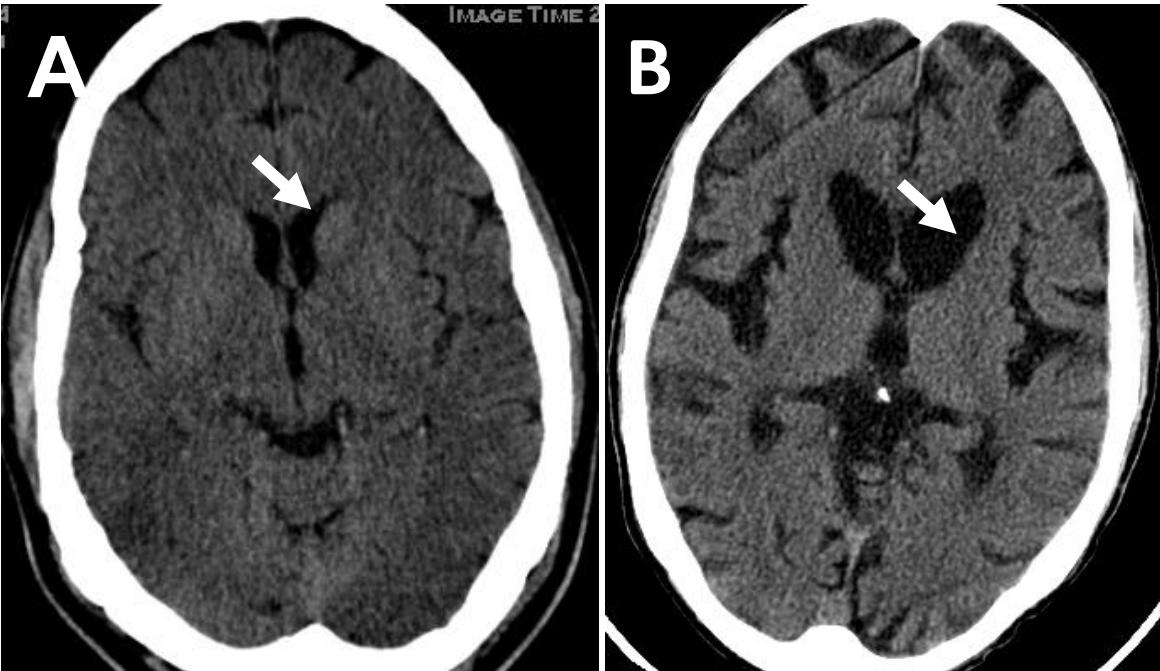
After 4-6 years...



After 10 years



Progressive cerebellar degeneration and flattening of the pons as the disorder progresses



Transverse brain computed tomography of (A.) a healthy person and (B.) a person with Huntington disease. The arrow shows atrophy of the head of the caudate nucleus, and generalized cortical atrophy is evident in the patient with Huntington disease.



The end

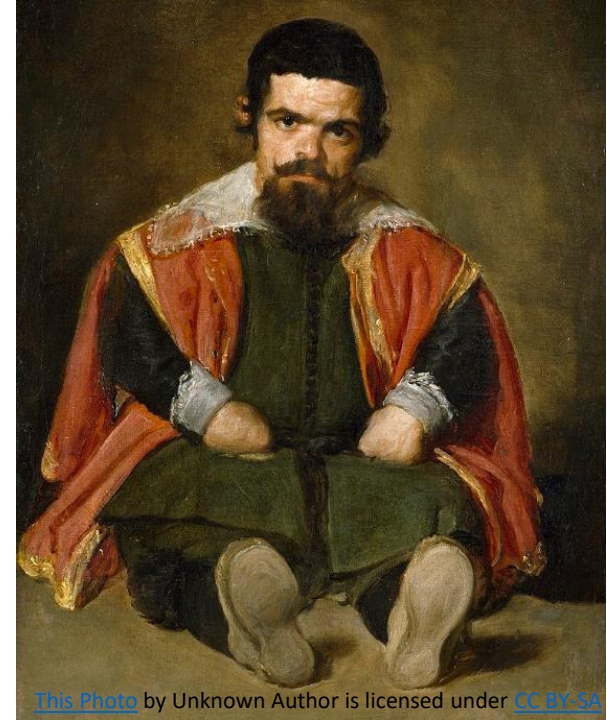
Unless there is more time, then we can do some more questions as
seen on the next x slides

Upon birth a male infant was recognized to have a genetic disorder caused by premature ossification of the growth plate during embryonic formation of the long bones. Genetic testing shows that a well-known single base pair change is responsible for the phenotype. Which of the following genetic event is most likely the that led to the disorder in this patient?

- | | |
|-----------------------|----|
| A. Anticipation | 0% |
| B. Heteroplasmy | 0% |
| 😊 C. Gain of function | 0% |
| D. Haploinsufficiency | 0% |
| E. Genetic drift | 0% |

Achondroplasia

- *FGFR3* mutations have been known for many centuries
- *FGFR3* codes for a transmembrane tyrosine kinase receptor that is involved in differentiation of cartilage to bone
- Specific Mutations in *FGFR3* may result in severe stunting of growth
- **Specific mutation** causes receptor to signal in an unregulated manner (that is the tyrosine kinase function occurs in the absence of hormone bound = “gain of function”)



Achondroplasia

A 7-year-old girl is brought to the physician by her parents for follow-up examination because of a 2-year history of progressively worsening ataxia. Genetic testing shows that she has Friedreich ataxia. Which of the following best explains the inheritance pattern of this disorder?

- 😊 A. Triplet repeat expansion occurs in an intron 0%
- B. The triplet repeat expansion in the 5' UTR causes other mRNA species to be incorrectly spliced 0%
- C. The repeat expansion causes the gene to be overexpressed 0%
- D. The repeat expansion causes gain of function because it occurs in an exon 0%
- 😊 E. The repeat expansion suppresses gene expression, but only one functional allele is needed so it is autosomal recessive 0%